

L₇₀ ratings. Most manufacturers suggest a lifetime of around 35,000 to 50,000 hours for their 7W LED retrofit lamps operating at 85V to 265V AC. However, some claim even higher values too. In general, white LEDs last 20-30 times longer than incandescent lamps, and 7-10 times longer than CFL lamps.

For more detailed information on how the useful life is calculated, you can refer the following websites:

http://apps1.eere.energy.gov/buildings/publications/pdfs/ssl/lifetime_white_leds.pdf

http://www.etaplighting.com/uploadedFiles/Downloadable_documentation/documentatie/whitepaper_LED_EN.pdf

<http://www.digikey.com/us/en/techzone/lighting/resources/articles/determining-led-rated-life.html>

QR Code Generator

I am facing problems in 'JavaScript based Quick Response Code Generator Interface' software project published in September issue. First, the QR code generated could not be saved. Also, it is giving error message "Stop! error encountered. Error: Bad filename or number." Please help!

Thiyaneswaran
Through e-mail

The author Sridhar Bukya replies:

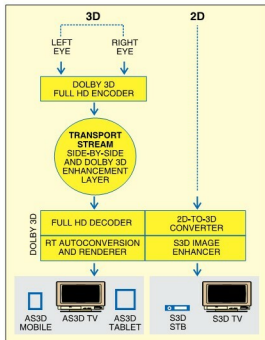
If you have downloaded the code from efymag.com website or EFY DVD, you cannot run the 'Save...' command from the compressed folder itself. You have to unzip it first, save the file into your hard disk, run the program, then save the QR code.

Also, you need to change the path in the source code defined under `HDD()` function. That is, the path in source code "`p="C:\paths\..."`" should be replaced with the correct path.

You can input any text of 60 characters length to generate the corresponding QR code.

Versatile Object Counter

I assembled the 'Versatile Object Counter' circuit published in July issue, which worked properly without



Dolby 3D technology (courtesy: Dolby)

extension circuit. When I connected the extension circuit to the counter, the counter counted from 0 to 9 but didn't work properly thereafter as mentioned in the text. IC CD4518 was automatically reset after count '9.'

Manohara G.

Through e-mail

EFY: Thanks for the interest in this circuit! The circuit was tested at EFY Lab and found working well. Note that there are four AND gates inside 7408 IC. Output pin 1 of the first gate is connected to master reset pin 7 of

CD4518 IC for resetting the display. If you are using the extension circuit, you should use the second AND gate N2. This AND gate comprises pins 4, 5 and 6 of 7408 with its output pin 6 connected to pin 7 of IC5.

Glasses-free 3D Technologies

I came to know of another interesting technology after completing my article on glasses-free 3D, which was published in November issue. For the benefit of readers, I am sharing some related information here:

This effective glasses-free 3D technology was jointly developed by Dolby and Philips, and launched in

April this year. Dolby 3D is a full high-definition 3D format and technology suite designed to enhance 3D experience across consumer devices, including TVs, tablets, smart phones and PCs. It automatically optimises 3D content for the specific device and screen size, upgrades half-resolution 3D to full resolution (for both glasses-free and traditional glasses-based 3D), and allows viewers to sit anywhere in the room by rendering multiple views for each display (unlike other technologies that provide only a few 'sweet spots' which viewers have to stick to).

Key components in this technology are:

1. A frame-compatible full-resolution (FCFR) video codec enhancement for all H.264 devices
2. Real-time (decoder-side) depth extraction (auto-conversion) and multi-view renderer
3. Depth adjustment for stereoscopic (S3D) and auto-stereoscopic (AS3D) displays, by user or automatically
4. State-of-the-art real-time 2D-to-3D conversion

The figure shown above describes the technology better.

Janani Gopalakrishnan Vikram
The author

ERRATA

❑ In 'Automatic Headlight Switcher' circuit in November issue, switch S2 for manual operation is wrongly connected across power supply. Instead of connecting S2 across the relay terminals, please connect it to the relay's normally-open terminal in series with the bulb.

❑ In 'Presence Sensing Lights Controller' construction project in November issue, there is a misprint in the source code (LIGHT.C). Instead of 'Control,' only 'ontrol' is printed.